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Started	Saturday, 10 October 2026, 3:26 PM
Completed	Saturday, 10 October 2026, 3:31 PM
Duration	4 mins 47 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Dividing the input into smaller subproblems
c. Comparing each element with the others
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **2**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470
b. 40,80,90,150,200,300,400,470,70,120
c. 80,150,200,470,40,90,300,400,70,120
d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **3**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Extraction
b. Insertion
c. Selection
d. Distribution

The correct answer is:
Insertion

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass
b. It allows for efficient parallel processing and sorting of data
c. It avoids the need for recursion
d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Removing duplicates from a list
b. Merging two datasets
c. Finding a specific element in a list
d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Quick Sort
b. Bubble Sort
c. Merge Sort
d. Heap Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Using the divide-and-conquer approach
b. Sorting data in a non-sequential manner
c. Bubbling up the largest element to its correct position with each pass
d. Being the most efficient sorting algorithm

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **8**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By sorting data in a single pass
b. By comparing elements sequentially
c. By breaking the input into smaller parts and merging them
d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Quick sort
b. Merge sort
c. Selection sort
d. Heap sort

The correct answer is:
Merge sort

Question **10**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. arrange()
b. sorted()
c. order()
d. sort()

The correct answer is: sorted()

Question **11**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It makes the analysis slower
b. It allows for quicker identification of duplicates
c. It has no effect on the analysis process
d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **12**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It bubbles up the largest element in each pass
- b. It uses the heap data structure
- c. It recursively sorts subproblems
- d. It divides the list into sublists

The correct answer is: It bubbles up the largest element in each pass

Question **13**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- b. A sorting algorithm is stable if it preserves the order of all keys
- c. A sorting algorithm is stable if it preserves the order of duplicate keys
- d. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **14**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It does not require recursion
- b. It is less efficient than bubble sort
- c. It is based on the divide-and-conquer approach
- d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **15**

Complete

Mark 0.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Linear Search
- b. Merge Sort
- c. Binary Search
- d. Bubble Sort

The correct answer is: Merge Sort